Import Required Libraries

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
from sklearn.preprocessing import LabelEncoder
import seaborn as sns
import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor
```

Load Dataset

In [2]:
```
df = pd.read_csv('Marketing_sales_data.csv')
df.head()
```

Out[2]:

	TV	Radio	Social Media	Influencer	Sales
0	Low	3.518070	2.293790	Micro	55.261284
1	Low	7.756876	2.572287	Mega	67.574904
2	High	20.348988	1.227180	Micro	272.250108
3	Medium	20.108487	2.728374	Mega	195.102176
4	High	31.653200	7.776978	Nano	273.960377

Exploratory Data Analysis

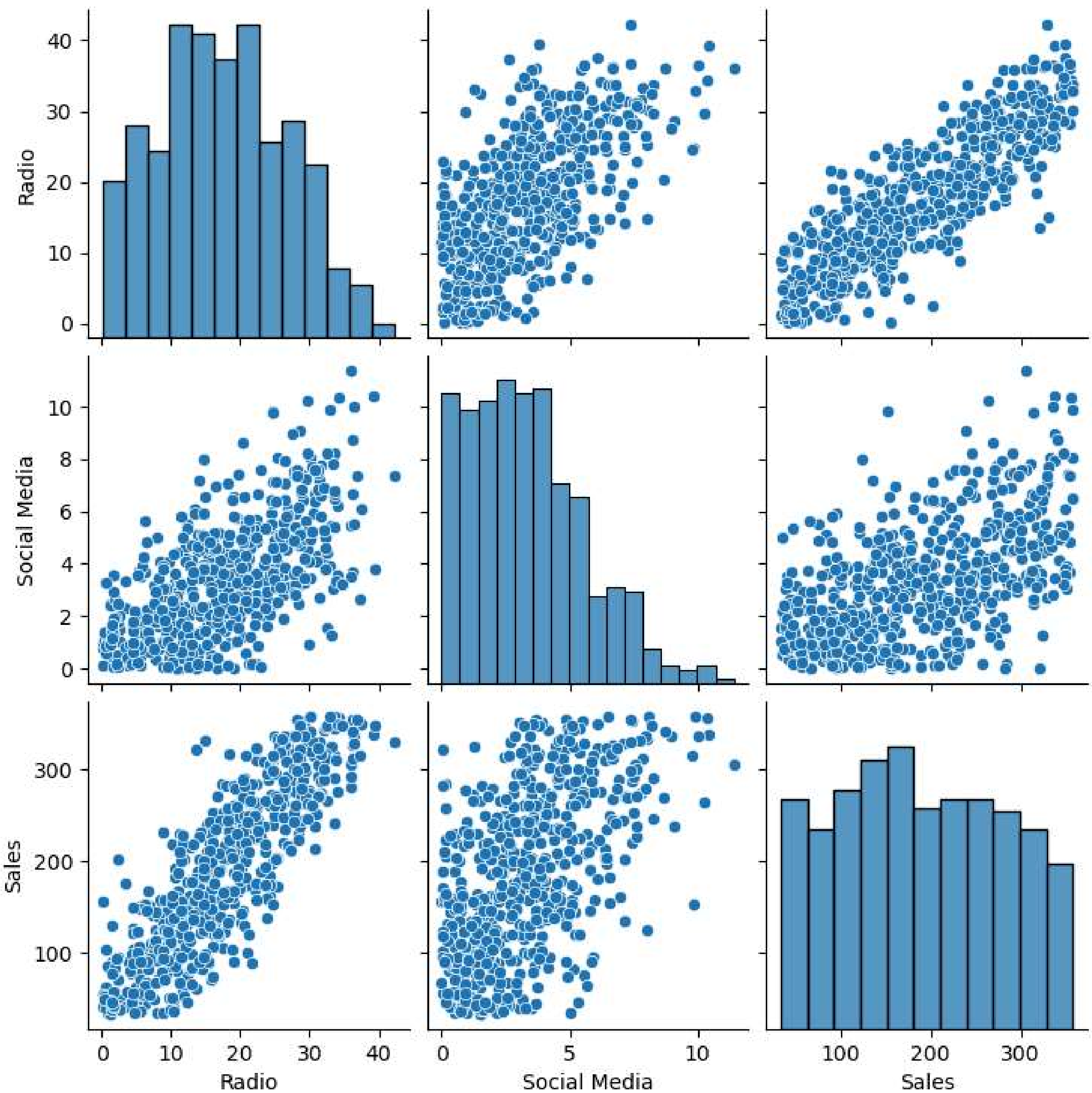
In [3]:
```
df.info()
df.describe()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 572 entries, 0 to 571
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   TV            572 non-null    object 
 1   Radio         572 non-null    float64
 2   Social Media  572 non-null    float64
 3   Influencer    572 non-null    object 
 4   Sales         572 non-null    float64
dtypes: float64(3), object(2)
memory usage: 22.5+ KB
```

Out[3]:
```
TV              0
Radio           0
Social Media    0
Influencer      0
Sales           0
dtype: int64
```

Visualize Relationships

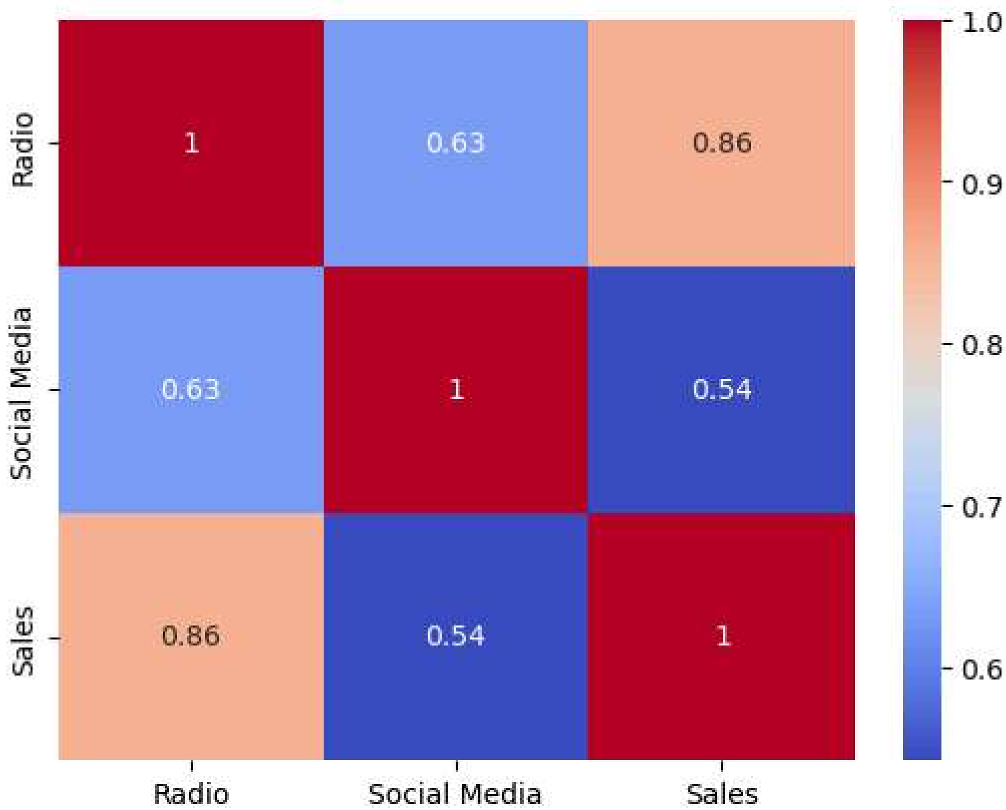
In [4]:
```
sns.pairplot(df)
plt.show()
```



Check Multicollinearity

In [5]:
```
## correlation matrix
```

```
numeric_df = df.select_dtypes(include=[np.number])
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')
plt.show()
```



Variance Inflation Factor

In [6]:
```
## using label encoder to convert categorical variables into numerical values
le_tv = LabelEncoder()
le_influencer = LabelEncoder()
df['TV'] = le_tv.fit_transform(df['TV'])
df["Influencer"] = le_influencer.fit_transform(df["Influencer"])
df.head()
```

Out[6]:

	TV	Radio	Social Media	Influencer	Sales
0	1	3.518070	2.293790	2	55.261284
1	1	7.756876	2.572287	1	67.574904
2	0	20.348988	1.227180	2	272.250108
3	2	20.108487	2.728374	1	195.102176
4	0	31.653200	7.776978	3	273.960377

In [7]:
```
## Using Variance Inflation Factor (VIF) to check for multicollinearity among the i
x=df[['TV', 'Influencer', 'Social Media']].copy()
x['Intercept'] = 1
vif = pd.DataFrame()
```

```
vif["Feature"] = x.columns
vif["VIF"] = [variance_inflation_factor(x.values, i) for i in range(x.shape[1])]
vif
```

Out[7]:

	Feature	VIF
0	TV	1.075837
1	Influencer	1.002260
2	Social Media	1.073787
3	Intercept	8.611724

Build Multiple Linear Regression Model

In [8]:
```
## Define x and y variables for regression analysis
x = df[['TV', 'Influencer', 'Social Media']]
y = df['Sales']
```

In [9]:
```
## Adding a constant to the independent variables for the regression model
x = sm.add_constant(x)
```

In [10]:
```
### Fitting the regression model
model = sm.OLS(y, x).fit()
model.summary()
```

Out[10]: OLS Regression Results

Dep. Variable:	Sales	**R-squared:**	0.376
Model:	OLS	**Adj. R-squared:**	0.373
Method:	Least Squares	**F-statistic:**	114.2
Date:	Mon, 15 Jun 2026	**Prob (F-statistic):**	6.95e-58
Time:	20:23:34	**Log-Likelihood:**	-3249.2
No. Observations:	572	**AIC:**	6506.
Df Residuals:	568	**BIC:**	6524.
Df Model:	3		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
const	161.9033	8.732	18.542	0.000	144.753	179.054
TV	-33.2603	3.846	-8.648	0.000	-40.814	-25.706
Influencer	0.5086	2.681	0.190	0.850	-4.757	5.774
Social Media	18.6311	1.379	13.514	0.000	15.923	21.339

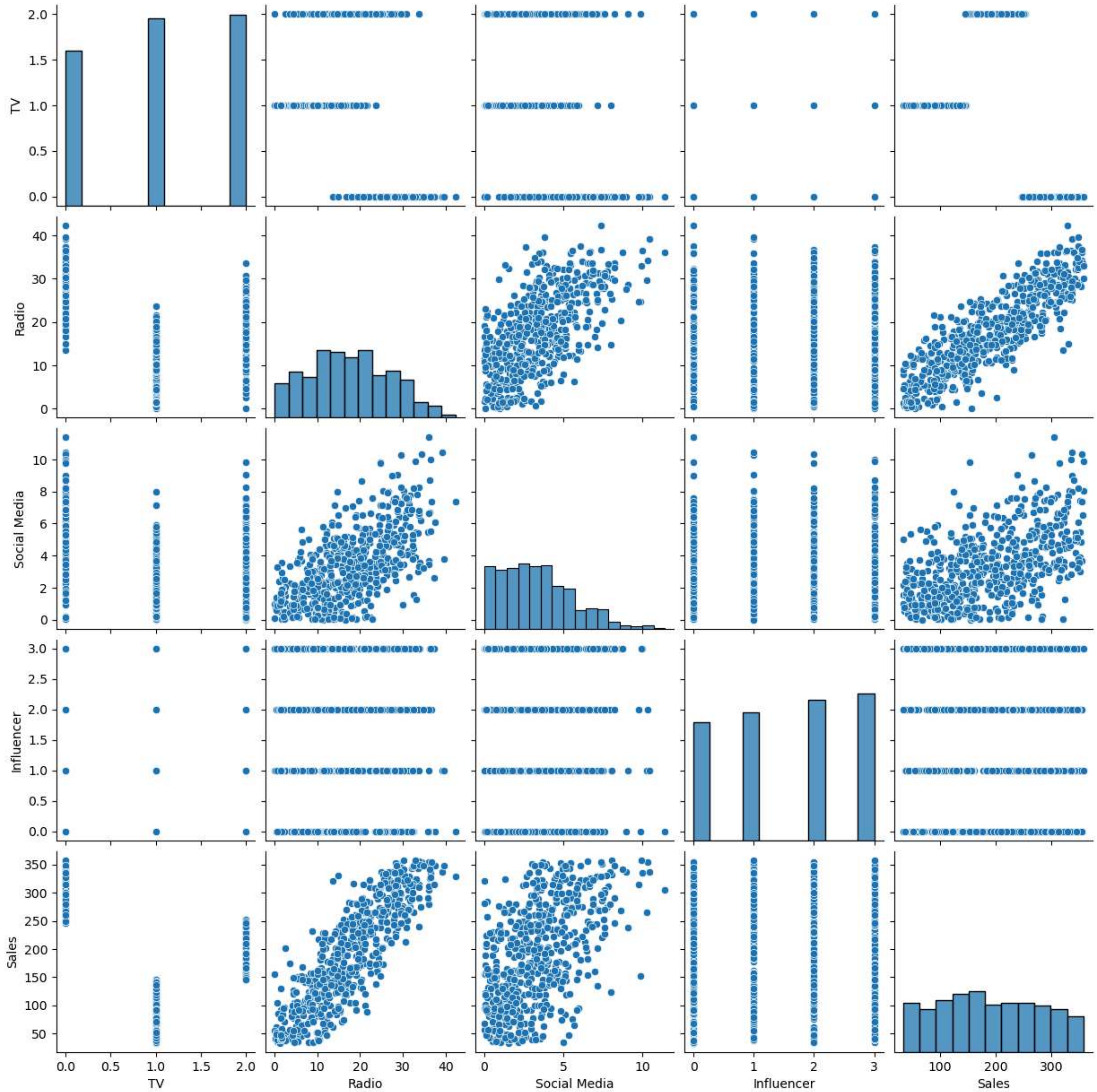
Omnibus:	43.649	**Durbin-Watson:**	1.915
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	21.091
Skew:	-0.281	**Prob(JB):**	2.63e-05
Kurtosis:	2.246	**Cond. No.**	13.7

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Diagnostic plots for regression analysis

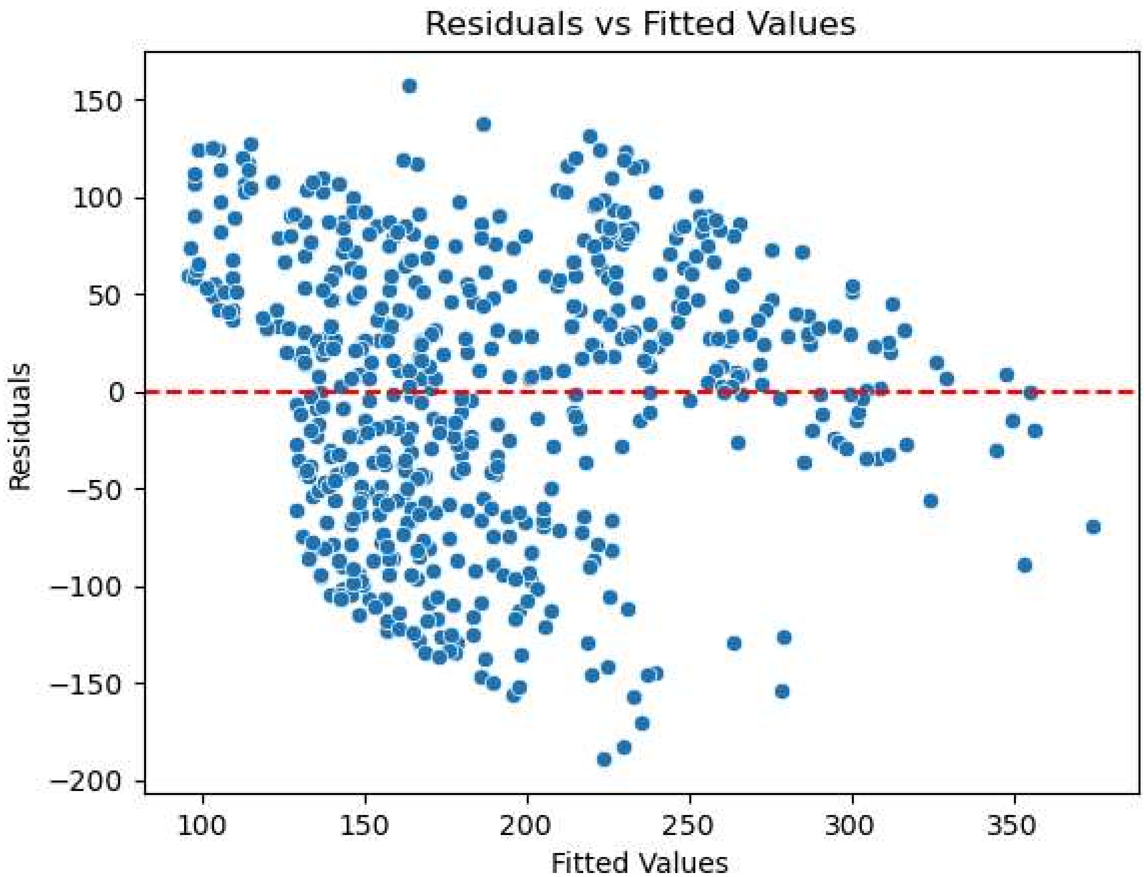
In [11]:
```
## Linearity
sns.pairplot(df)
```

Out[11]: `<seaborn.axisgrid.PairGrid at 0x25e20e94a50>`

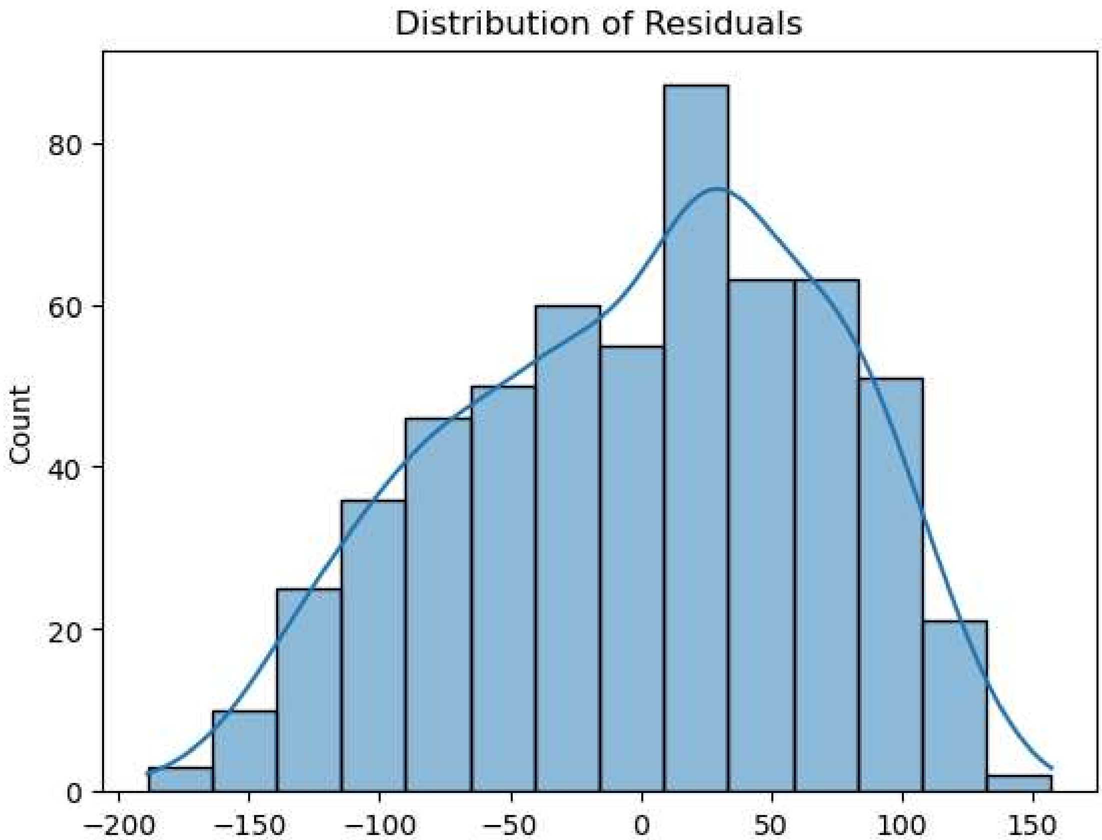


In [12]:
```
## Residual analysis
residuals = model.resid
fitted_values = model.fittedvalues
sns.scatterplot(x=fitted_values, y=residuals)
plt.axhline(0, color='red', linestyle='--')
plt.xlabel('Fitted Values')
plt.ylabel('Residuals')
plt.title('Residuals vs Fitted Values')
```

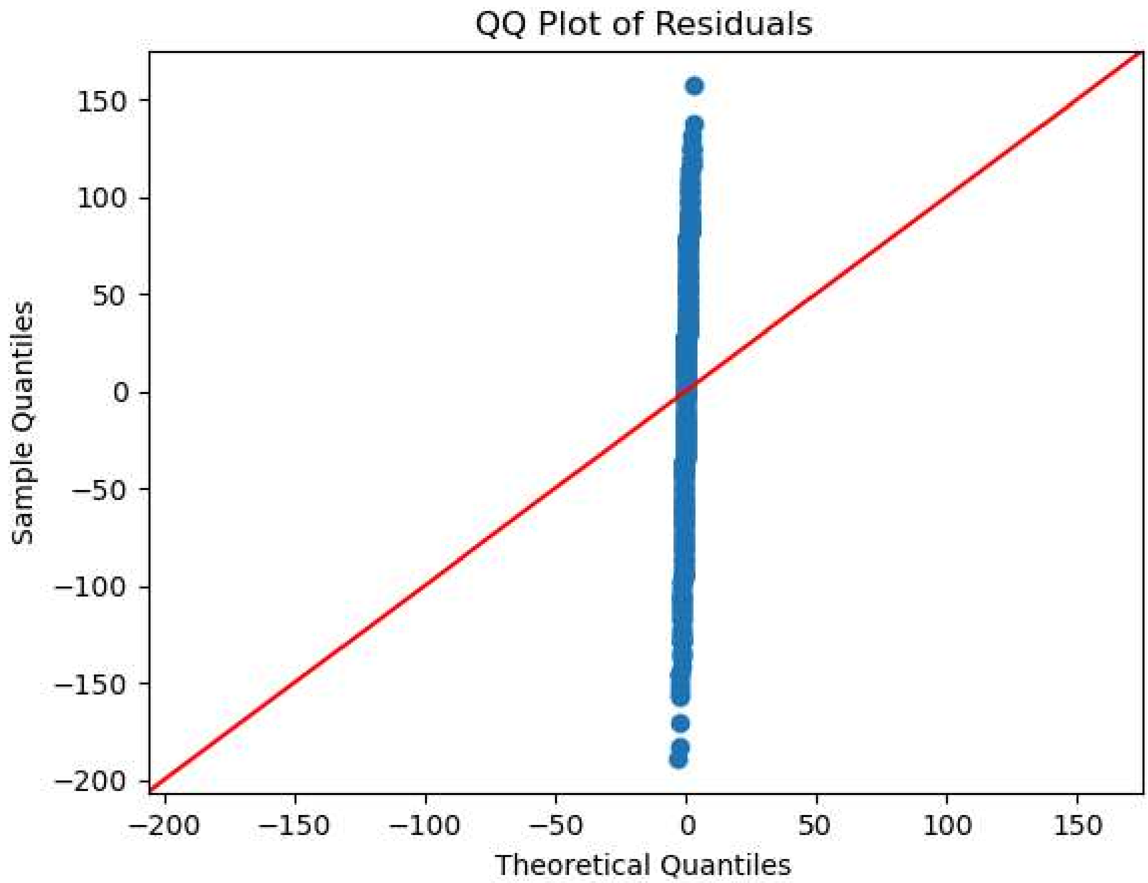
Out[12]: Text(0.5, 1.0, 'Residuals vs Fitted Values')



In [13]:
```
### Normality of residuals
sns.histplot(residuals, kde=True)
plt.title('Distribution of Residuals')
plt.show()
```



In [14]:
```
# QQ plot for normality
sm.qqplot(residuals, line='45')
plt.title('QQ Plot of Residuals')
plt.show()
```



In [15]:
```python
## To check the coefficients of the model
print(model.params)
```

```
const           161.903269
TV              -33.260271
Influencer        0.508635
Social Media     18.631111
dtype: float64
```

Coefficient Interpretation (Business Meaning)

Your model: Scales =161.903269−33.260271(TV)+0.508635(Influencer)+18.631111(Social Media)

`TV (-33.260271)`

Holding Influencer and Social Media constant, every 1-unit increase in TV advertising spend is associated with a decrease of approximately 33.26 units in Sales.

`Influencer (0.508635)`

Holding other channels constant, every 1-unit increase in Influencer marketing spend increases Sales by approximately 0.51 units.

Social Media (18.631111)

Holding other channels constant, every 1-unit increase in Social Media spending increases Sales by approximately 18.63 units.

Intercept (161.903269)

When all marketing spends are zero, baseline Sales is approximately 161.9 units.

Business Interpretation

In [16]:
```
## Business Interpretation of Results.

print(f"Coefficient for TV Advertising: {model.params['TV']}")
print(f"P-Values: {model.pvalues}")
print(f"R-squared: {model.rsquared}")
```

```
Coefficient for TV Advertising: -33.26027132218452
P-Values: const           2.282195e-60
TV              5.361789e-17
Influencer      8.495844e-01
Social Media    2.782559e-36
dtype: float64
R-squared: 0.3762985299010331
```

Business Recommendation

Based on the regression analysis, marketing budget allocation should prioritize Social Media advertising, as it shows the strongest positive impact on Sales. Influencer marketing should be maintained at a moderate level as a supporting channel. However, TV advertising appears to have a negative relationship with Sales, suggesting inefficiencies in targeting or spending strategy. The company should reconsider or restructure TV campaigns before further investment.

In [ ]: